TIC 257923571

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

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BLS EPHEMERIS
  Period       0.885246 d  (ALIASED, see below)
    -> corrected period: 1.770493 d
  Epoch (t0)   2448.4166 BTJD
  Duration     0.0840 d
  Depth        0.942 %  (9,416 ppm)
  SNR          199.7

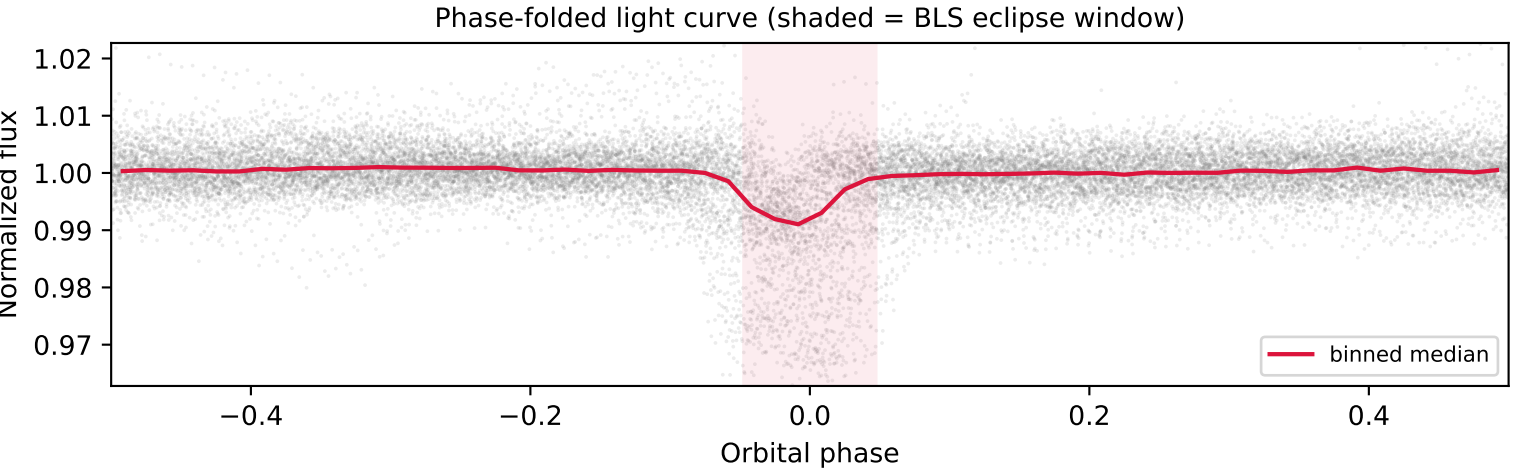
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     93  (gate >= 3)
  Robust depth ratio    0.70  (gate >= 0.50)
  Robust depth sigma    69.3  (gate >= 7.0)
  Odd/even z-score      26.90  (ALIAS FLAGGED)
  Secondary eclipse     not detected
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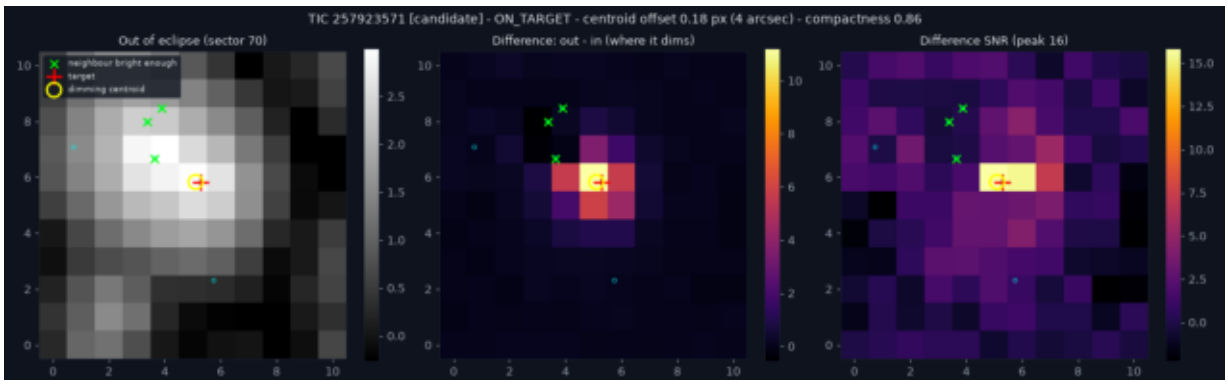
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 70    light-curve dip in this sector: 73.8 sigma    t0 refined by 0.0000 phase
  Difference-image peak SNR   15.83   (gate >= 5.0)      Compactness   0.86   (gate >= 0.50)
  Dimming centroid offset     0.18 px (4 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 257923569
    1.68 px away, dTmag -0.39 - farther from the dimming than the target
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Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 26.90 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 1.770493 d, in any write-up.